

NORRBERG-NORRHOG V, Sweden

WMO: 023370

Lat: 62.2632N

Lon: 15.6709E

Elev: 1050

StdP: 14.15

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.4	-12.9	-23.2	1.6	-18.2	-17.4	2.2	-12.8	11.0	31.0	9.6	28.5	0.2	320	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	79.7	60.3	75.7	59.3	71.6	57.6	63.5	73.5	61.4	70.2	59.5	67.7	2.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.1	80.6	65.7	58.2	75.2	63.8	56.1	69.8	61.7	29.4	73.6	27.9	70.5	26.5	67.8	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.2	7.7	DB	-21.6	85.3	5.7	3.2	-25.7	87.6	-29.0	89.4	-32.2	91.2	-36.3	93.6	(4)
(5)			WB	-22.3	66.8	5.1	0.9	-25.9	67.5	-28.9	68.0	-31.8	68.5	-35.4	69.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	15.0	18.0	26.3	36.6	45.1	53.0	58.3	55.3	46.2	36.0	26.9	18.6		
	DBStd	17.02	12.90	11.31	8.99	5.90	6.23	5.84	5.38	5.42	6.10	7.10	9.00	13.26		
	HDD50	5585	1084	896	735	401	178	32	3	14	142	433	692	975		
	HDD65	10454	1549	1316	1200	851	615	361	217	302	563	898	1142	1440		
	CDD50	619	0	0	0	1	28	122	260	179	29	0	0	0		
	CDD65	13	0	0	0	0	0	1	10	2	0	0	0	0		
	CDH74	497	0	0	0	0	21	158	242	76	0	0	0	0		
	CDH80	80	0	0	0	0	2	23	45	10	0	0	0	0		
	WSAvg	2.6	2.3	2.6	3.3	3.2	3.6	2.9	2.2	1.7	2.1	2.6	2.5	2.1		
	PrecAvg	24.0	1.5	1.1	1.1	1.4	1.8	2.6	3.5	3.2	2.1	2.2	1.9	1.7		
(15) Precipitation	PrecMax	34.6	2.8	2.0	2.5	4.4	4.3	4.8	8.0	5.4	5.6	5.5	4.8	2.6		
	PrecMin	19.3	0.2	0.1	0.2	0.3	0.6	1.1	1.1	1.8	0.6	0.7	0.7	0.2		
	PrecStd	3.6	0.7	0.5	0.5	1.0	1.0	0.9	1.7	1.1	1.2	1.2	1.1	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	42.9	51.6	66.2	76.9	82.9	84.6	81.0	69.1	56.7	49.0	42.8		
		MCWB	36.1	37.2	41.4	50.5	53.5	58.7	63.9	62.8	56.2	49.8	46.0	38.8		
	2%	DB	36.8	39.1	46.2	59.0	69.8	78.9	80.3	75.9	64.1	53.0	44.9	39.8		
		MCWB	34.0	35.2	38.2	46.2	52.7	58.5	61.9	61.0	54.0	48.1	42.4	36.8		
	5%	DB	34.1	36.1	42.7	54.8	64.3	73.8	76.2	71.6	60.4	50.0	41.2	37.8		
		MCWB	32.4	33.1	36.4	43.5	50.0	56.4	60.1	59.3	52.4	46.4	39.5	35.3		
	10%	DB	31.8	33.1	39.4	50.5	59.3	68.1	72.1	67.6	57.1	47.5	38.4	35.0		
		MCWB	30.5	30.9	34.4	41.2	48.4	53.9	58.7	57.8	50.8	44.9	36.9	33.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.8	37.6	42.9	52.2	56.9	61.9	66.3	65.5	58.3	51.5	46.1	39.5		
		MCDB	40.0	41.1	50.6	61.2	70.8	75.4	79.7	74.0	64.7	54.7	49.0	41.4		
	2%	WB	34.6	35.7	38.7	47.7	54.6	59.6	64.1	63.5	55.7	49.0	42.6	37.5		
		MCDB	36.6	39.0	44.6	57.9	65.7	73.2	76.0	71.6	60.6	51.4	44.6	39.4		
	5%	WB	32.5	33.3	36.9	44.3	51.8	57.4	62.1	61.2	53.9	47.2	39.8	35.4		
		MCDB	33.7	35.5	42.0	52.8	62.0	71.0	71.5	68.3	58.7	49.6	41.0	37.4		
	10%	WB	30.5	31.1	34.8	41.7	49.5	55.3	60.1	59.3	52.1	44.8	37.2	33.1		
		MCDB	31.7	33.0	38.9	49.5	58.1	65.7	68.6	65.3	56.0	47.5	38.4	34.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.5	17.4	20.4	21.1	22.3	23.7	22.9	21.9	19.4	15.2	11.8	13.9		
		MCDBR	12.9	15.4	22.5	31.9	33.1	36.4	32.2	30.5	25.1	19.3	13.6	12.9		
		MCWBR	11.5	12.0	15.2	18.9	17.0	16.2	14.9	15.9	15.4	14.0	11.8	11.1		
	5% WB	MCDBR	12.7	14.4	20.8	28.7	27.8	30.6	27.7	24.4	21.0	17.7	12.6	12.7		
		MCWBR	11.6	11.7	14.5	17.5	15.6	14.4	14.3	14.0	13.9	13.8	11.4	11.4		
(40) Clear-Sky Solar Irradiance	taub		0.222	0.277	0.303	0.338	0.343	0.338	0.354	0.346	0.314	0.299	0.238	0.344		
	taud		2.111	2.264	2.338	2.400	2.451	2.493	2.460	2.477	2.555	2.477	2.188	2.723		
	Ebn at Noon		150	219	258	270	278	282	273	266	256	214	141	98		
	Edh at Noon		10	19	26	30	31	30	30	28	21	16	10	6		
(44) All-Sky Solar Radiation	RadAvg		70	272	742	1235	1486	1700	1548	1183	747	334	97	37		
	RadStd		8	42	73	105	153	128	185	104	91	54	20	3		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page